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$$\textcircled{a} \quad \frac{K_{N,0}}{L_{N,0}} = 10 \quad \frac{Y_{N,0}}{L_{N,0}} = \frac{4 \cdot (100)^{\frac{1}{4}} \cdot (10)^{\frac{3}{4}}}{10} = 7,113$$

$$\frac{K_{S,0}}{L_{S,0}} = 400 \quad \frac{Y_{S,0}}{L_{S,0}} = \frac{5 \cdot (8000)^{\frac{1}{2}} \cdot (20)^{\frac{1}{2}}}{20} = 100$$

$$\textcircled{b} \quad \tilde{K}_i = \left[\frac{s_i A_i}{\delta + n} \right]^{\frac{1}{1-\alpha_i}} \quad N \begin{cases} K_N^* = 16 \\ Y_N^* = 8 \end{cases}$$

$$S \begin{cases} K_S^* = 625 \\ Y_S^* = 125 \end{cases}$$

Predictions:

- Relative converge
- Higher growth in $N \rightarrow$ due to $n_N > n_S$

③

$$k = 8100$$

$$L = 30$$

$$\alpha = \frac{1}{2}$$

$$A = 5$$

} Adopts South
Production function

$$k_{u,0} = 270 \rightarrow y_{u,0} = 82,16$$

$$y_{u,0} - y_{N,0} = 75 \rightarrow \text{Higher } A \quad \text{More capital and better use}$$

$$y_{u,0} - y_{s,0} = -17,84 \rightarrow k_{s,0} < k_{u,0}$$

Capital gets diluted at uni

South gains in L

$$\text{Before: } y_{N,0} + y_{s,0} = 2071,31$$

$$\text{after } y_{u,0} = 2464,752 \rightarrow \text{Not per capita (for S)} \\ \text{but still a gain}$$

$$\Delta Y = 393,62 \text{ or } 19\%$$

$$\textcircled{d} \quad n_u = 0,01 \cdot \frac{20}{30} + 0,03 \cdot \frac{10}{30} = \frac{4/3}{100} \approx 0,016667$$

$$k_u^* = 506,25 \rightarrow y_u^* = 112,5$$

* Slightly lower level, but higher growth

②

$$(a) \quad k = k_p + \overbrace{\Theta \cdot k_p}^{k_s} \rightarrow k_p = \frac{k}{1+\Theta}$$

$$\frac{k_s}{k} = \frac{\Theta \cdot k_p}{k_p(1+\Theta)} = \frac{\Theta}{1+\Theta} \quad \text{with } \Theta = \frac{1}{2} \leadsto \frac{k_s}{k} = \frac{1}{3} \text{ Lat } k$$

$$Y = A \left(\frac{k}{1+\Theta} \right)^\alpha \cdot L^{1-\alpha} \rightarrow Y = A \cdot \boxed{(1+\Theta)^{-\alpha}} \cdot k^\alpha L^{1-\alpha}$$

$$\left. \begin{aligned} Y^{NC} &= A \cdot k^\alpha \cdot L^{1-\alpha} \\ Y^C &= A \cdot (1+\Theta)^{-\alpha} \cdot k^\alpha \cdot L^{1-\alpha} \end{aligned} \right\} \frac{Y^C}{Y^{NC}} - 1 = (1+\Theta)^{-\alpha} - 1 \approx 12,64\%$$

Careful with
chosen base!

③

$$\max_{k,L} \quad \Pi = 1 \cdot \underbrace{A \cdot (1+\Theta)^{-\alpha}}_{\text{wedge}} \cdot k^\alpha \cdot L^{1-\alpha} - r \cdot k - w \cdot L$$

F.O.C:

$$\Pi_k : \underbrace{\frac{A}{(1+\Theta)^\alpha} \cdot \alpha \left(\frac{k}{L} \right)^{\alpha-1}}_{MPK} = r \quad \Pi_L : \underbrace{\frac{A}{(1+\Theta)^\alpha} \cdot (1-\alpha) \left(\frac{k}{L} \right)^\alpha}_{MPL} = w$$

Workers also
affected

Factor shares:

$$\frac{rK}{Y} = \frac{A \cdot (1+\theta)^{-\alpha} \cdot \alpha \cdot \left(\frac{K}{L}\right)^{\alpha-1}}{A(1+\theta)^{-\alpha} \cdot K^{\alpha} \cdot L^{1-\alpha}} \quad K = \alpha$$

$$\frac{w \cdot L}{Y} = 1 - \alpha$$

$$\textcircled{c} \quad Y = A \cdot \left(\frac{K}{1+\theta}\right)^{\alpha} L^{1-\alpha} \rightarrow Y = \boxed{\frac{A}{(1+\theta)^{\alpha}}} K^{\alpha} \cdot L^{1-\alpha}$$

$\hookrightarrow \hat{A}$

$\hat{A}$ gets distorted by non-observability of the type of capital

$\hat{\alpha} = \alpha \rightarrow$ No change since $\frac{r \cdot K}{Y}$ is directly observable

①

$$\Theta \rightarrow \frac{\Theta}{4}$$

- From growth rates:

$$g_Y = \alpha \cdot g_K + (1-\alpha) \cdot g_L + \underbrace{g_{Solow}}_{TFP + \Delta\Theta}$$

- Assuming A grows at rate g_A

$$\text{Ten-year transition: } \underbrace{\frac{1}{10}}_{\text{time period}} \cdot \underbrace{\ln\left(\frac{1+\Theta'}{1+\Theta}\right)}_{\text{log change over period}}$$

$$\therefore g_{Solow} = g_A + \frac{\alpha}{10} \cdot \ln\left(\frac{1+\Theta}{1+\Theta/4}\right)$$

with $g_A = 0$ and $\alpha = \frac{1}{3}$, $\Theta = \frac{1}{2} \rightarrow \Theta' = \frac{1}{8}$
we have:

$$g_{Solow} \approx 0,96 \text{ p.p. / year}$$

Worse crime $\rightarrow$ reverse g